Friedrich-Alexander-Universität
Technische Fakultät







# AI in Medicine

## 人工智能在医学中的应用

**Prof. Dr.-Ing. habil. Andreas Maier, Pattern Recognition Lab**

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany 30 October 2026 ▪ Shanghai Chest Hospital ▪ 上海市胸科医院

# About the Speaker

## Prof. Dr.-Ing. Andreas Maier 主讲人简介

### Responsibilities & Experience

- Head of the Pattern Recognition Lab, FAU
- Head of the Competence Center Medical Engineering Erlangen
- Head of the Medical AI Hub of Bavaria
- Member, AI Council of the Free State of Bavaria
- Scientific Counselor, Fraunhofer Institute
- AI Council, German Radiation Protection Society
- European Research Council (grantee & reviewer)
- Visiting Professor at Stanford University and MIT

### Industry & Entrepreneurship

- Siemens Healthineers (innovation manager / tech lead)
- Start-ups: *Mira Vision* (shareholder, valued at EUR 10 M); *Norizon* (council, early VC phase)
- Holds more than 15 active US and EU patents

*Theory first, application always.*



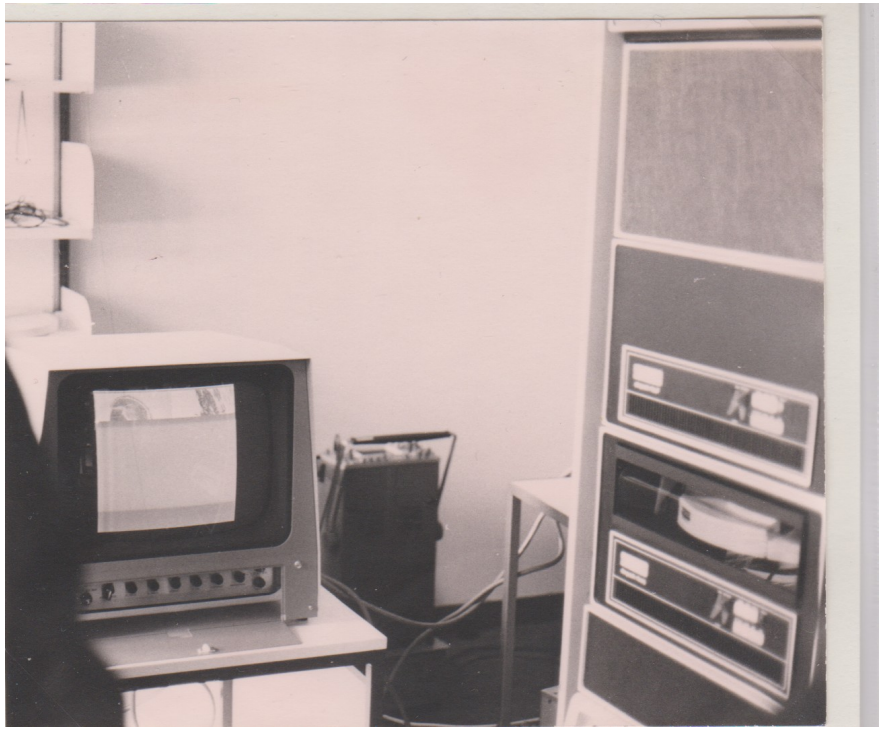
# Pattern Recognition Lab

Friedrich-Alexander-Universität Erlangen-Nürnberg 模式识别实验室

# Know the Past to Understand the Future

温故而知新

- Founded in 1975, founder: Heinrich Niemann
- Germany's first dedicated Pattern Recognition / Machine Learning Lab
- Medical imaging as early as 1981
- First computer: PDP-11/34 with 248 kB, 2.5 MHz
- Focus: theory & good understanding of hardware



# Our Team Today

## Pattern Recognition Lab 我们的团队

### Secretaries



Alexandra H.



Claudia S.

### Head of the Lab



Prof. Dr.-Ing. Andreas Maier

### Technician



Sven G.

- ∼60 PhD students & 8 postdocs
- 2 emeritus professors
- ∼3 709 master & bachelor students

- Academic partners in Europe, Asia, North/South America, Australia
- Industry partners: Siemens, Fraunhofer, Adidas, Intel, Audi, IBM, and more

# Research Areas

## Pattern Recognition Lab 研究方向

**Computer Vision**

**Time Series Intelligence**

**Magnetic Resonance Imaging**

**Data Processing for Utility Infrastructure**

**Learning Algorithms for Medical Big Data Analysis**

**Speech Processing & Language Understanding**

**Inverse Problems & Applications**

# Work at the Lab

## 实验室的工作







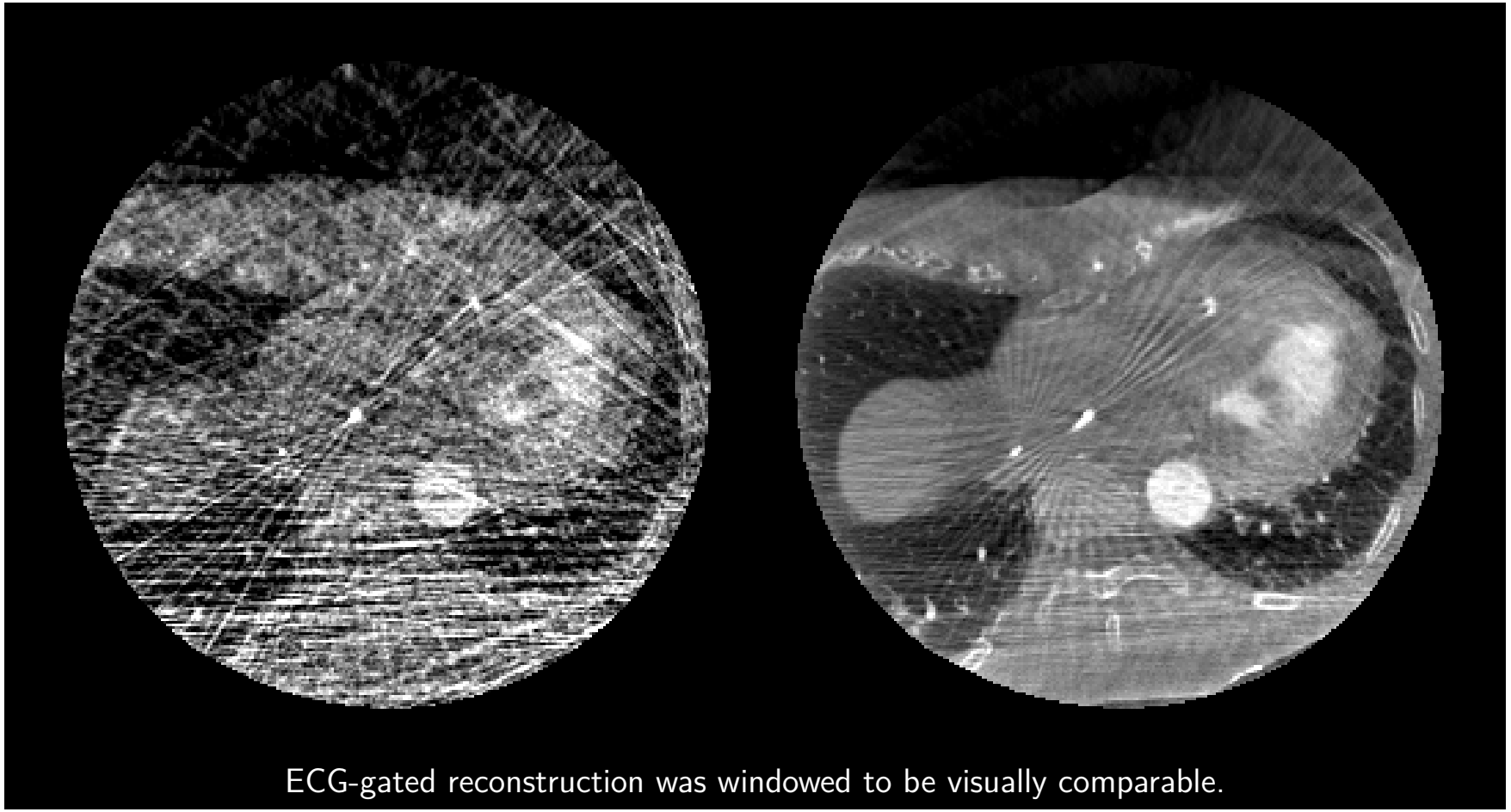







# Motion Compensation

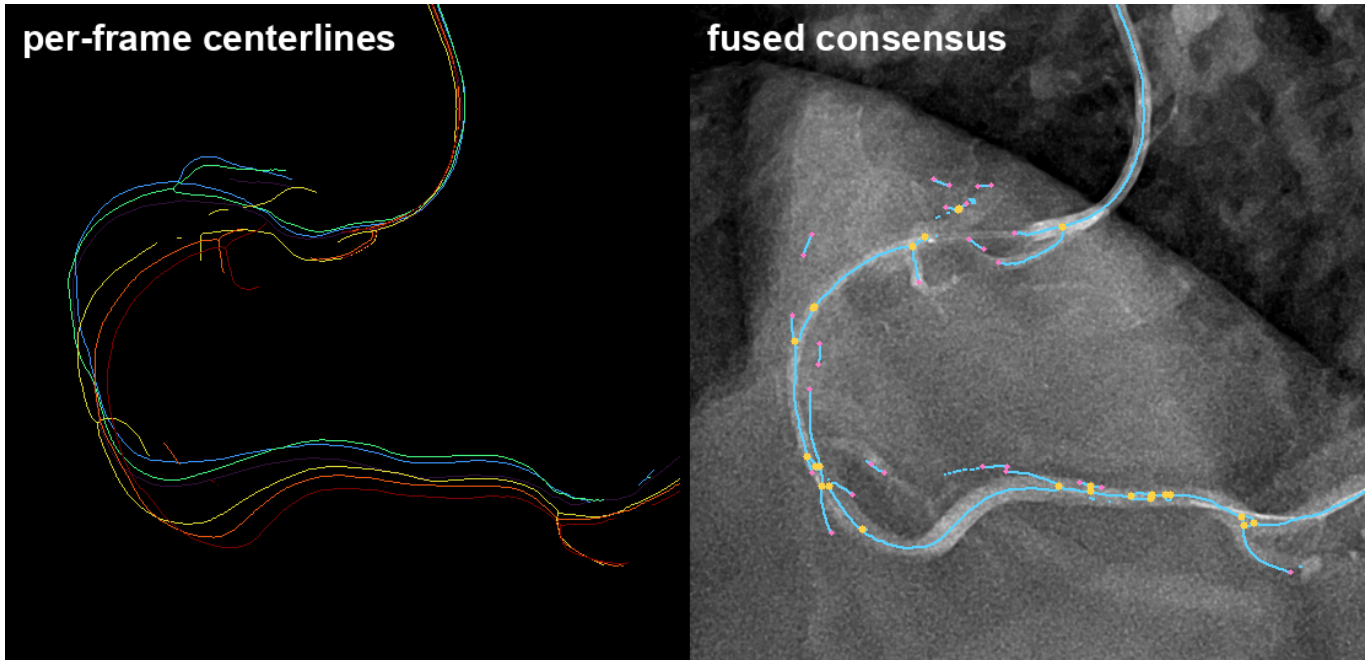
## 运动补偿



ECG-gated reconstruction was windowed to be visually comparable.

# VesselROVER

## Coronary centerlines in angiography 冠状动脉中心线提取




Left: per-frame centerlines of one run, spread by cardiac motion. Right: fused consensus centerline and branch points on a clinical frame.

- No single frame shows the complete coronary tree.
- Fuse several segmenters (STAPLE) *and* many frames (ROVER, from speech recognition).
- 13 patients, ∼300 clinical runs: centerline tree **1.22**× more complete, **6.7**× more distal branches.
- Open source: `github.com/akmaier/VesselROVER`

**Joint work with Dr. Wenzheng Han, Shanghai Chest Hospital**

A. K. Maier, S. Bayer, W. Han, *Combining rater fusion and time fusion for coronary centerline extraction from single-plane angiography*, SPIE Medical Imaging 2027 (accepted).

# Outline

## 提纲

- What is deep learning?
- What is agentic AI?
- What does it mean for science?

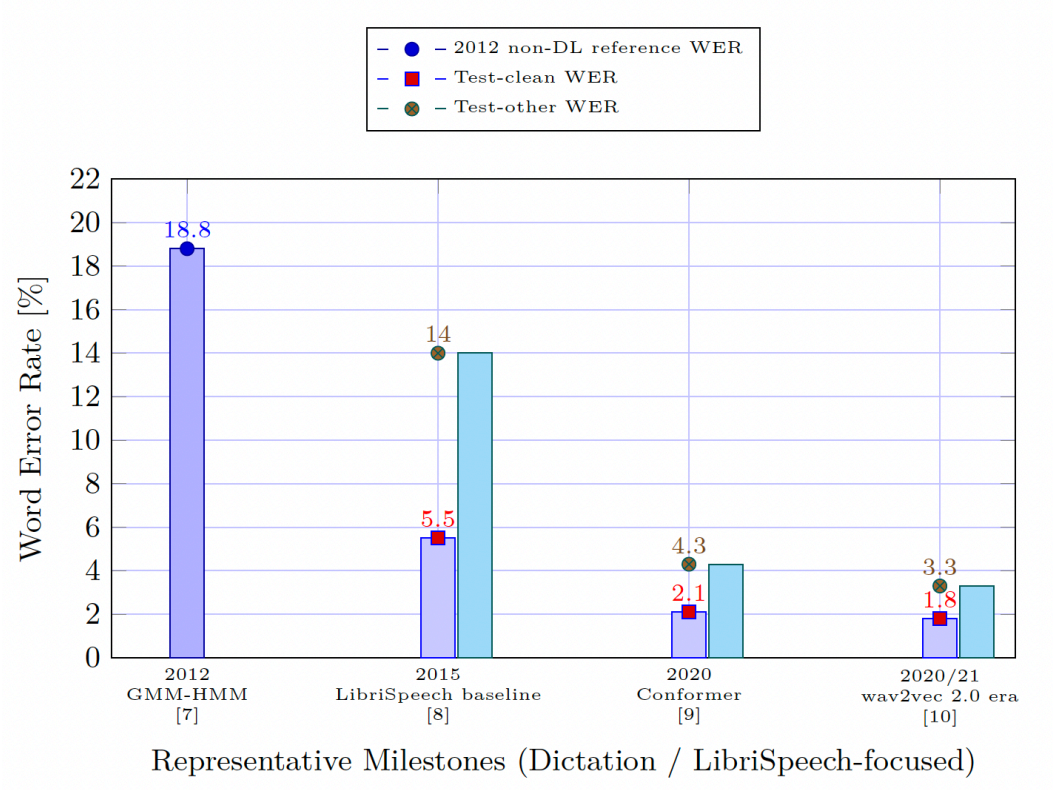
# What is deep learning?

什么是深度学习?

# Deep learning

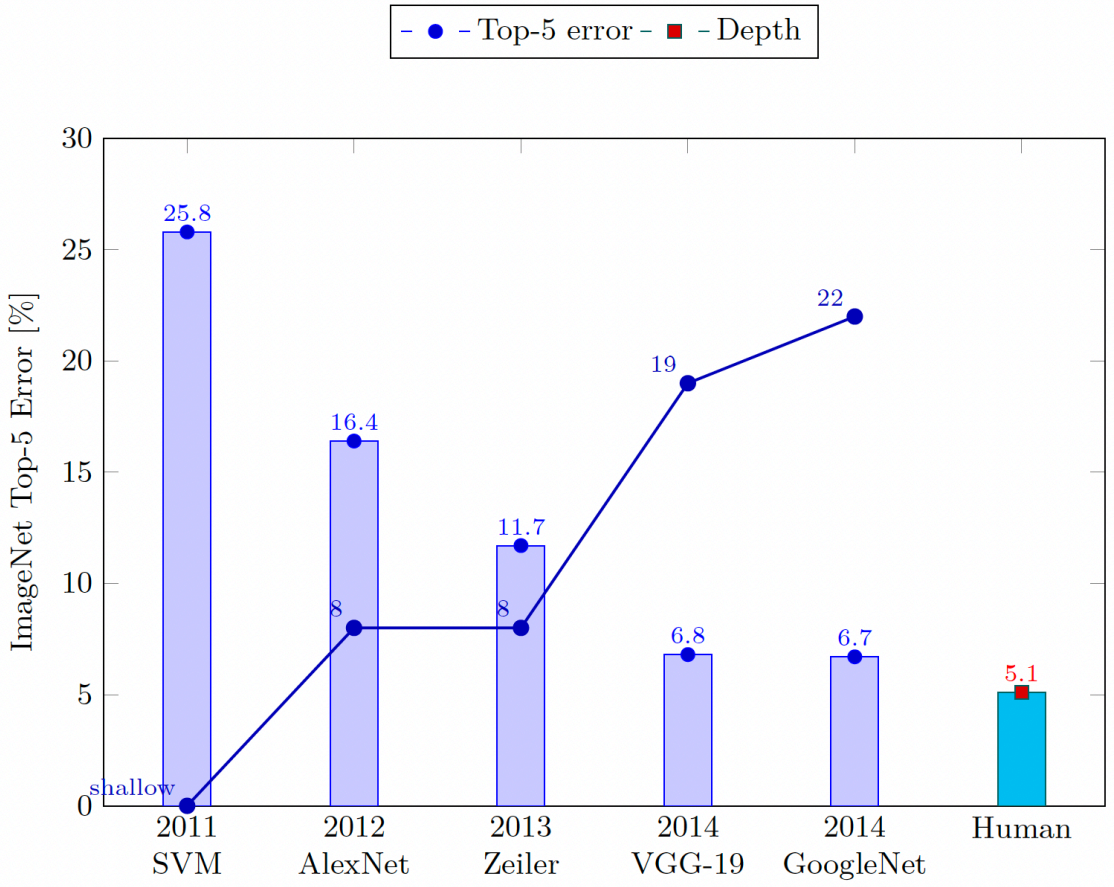
## Speech 深度学习

- Revolutionising pattern recognition and AI for more than 10 years.



# Deep learning

## Image classification 深度学习



# Deep learning

## Learning to play 深度学习



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.

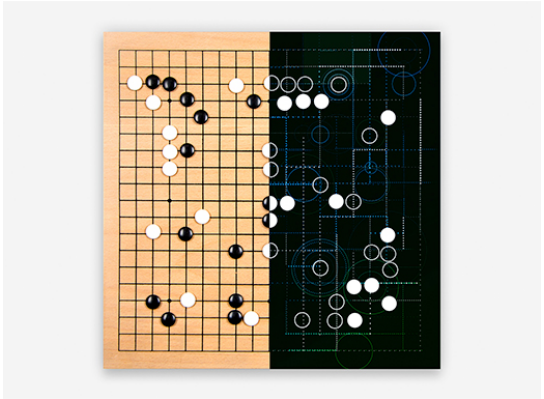
# Deep learning

## Learning to play 深度学习



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.



AlphaGo (DeepMind, 2016)

- Self-play + Monte-Carlo tree search.
- Defeats world champion Lee Sedol 4–1.

# Generative AI

## Source image 生成式人工智能



# Generative AI

## Image description 生成式人工智能




*"An athletically built man with a serious expression stands in a well-lit room with a wooden ceiling. He is wearing a blue t-shirt with the phrase 'SAVE THE CHUBBY UNICORNS' and an image of a rhinoceros. The man is taking a selfie with a modern smartphone, capturing his reflection in a mirror. In the background, one can see gym equipment including dumbbells and a weight rack. There is also a hanging chandelier with a unique design, illuminating the room."*

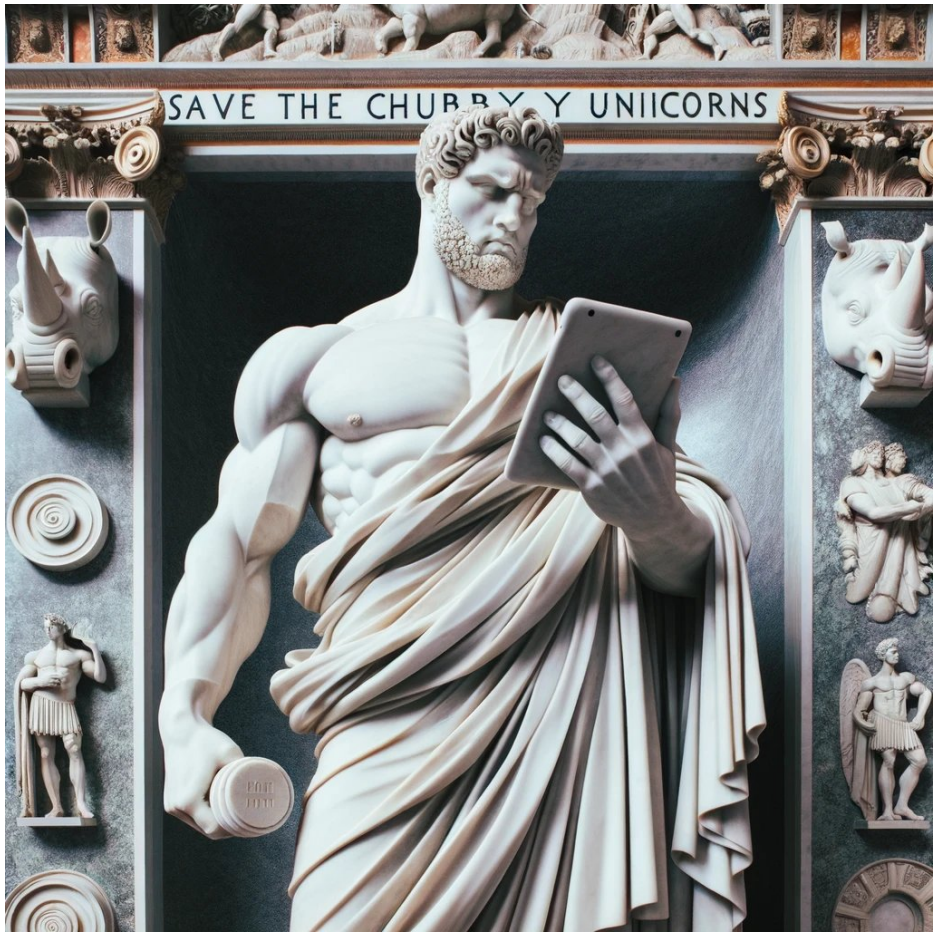
# Generative AI

## Generated image 生成式人工智能



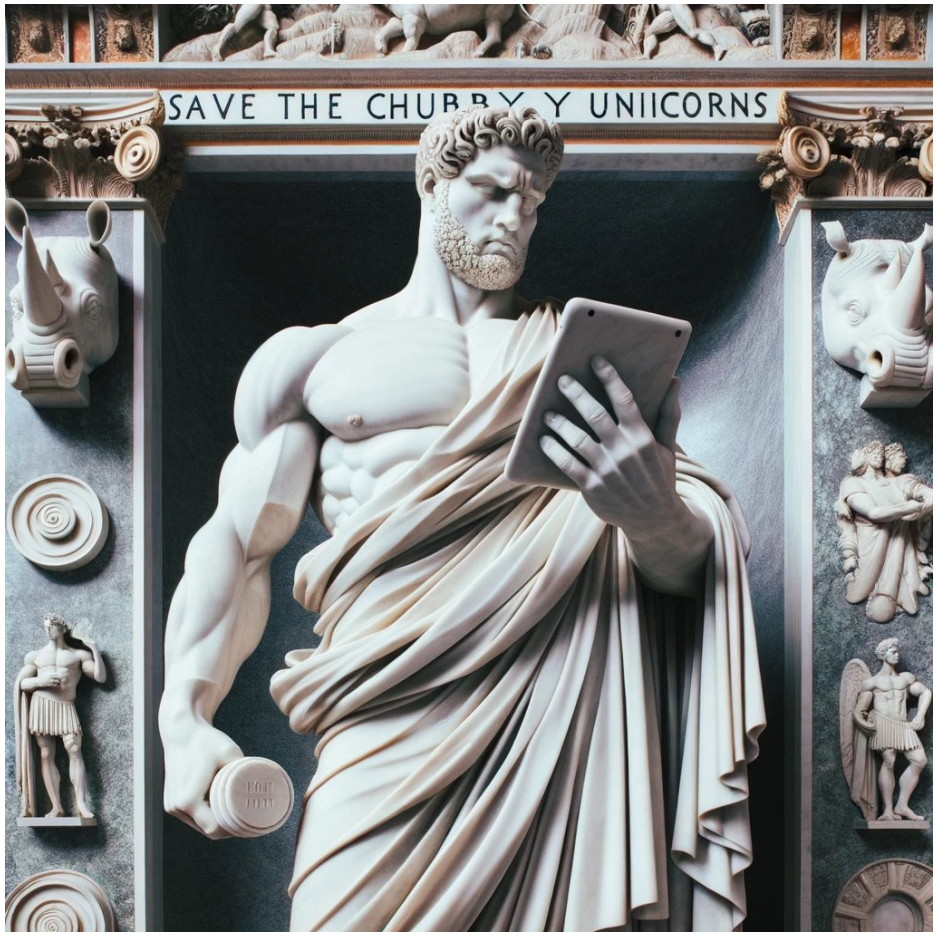
# Generative AI

## Style variations 生成式人工智能



# Generative AI

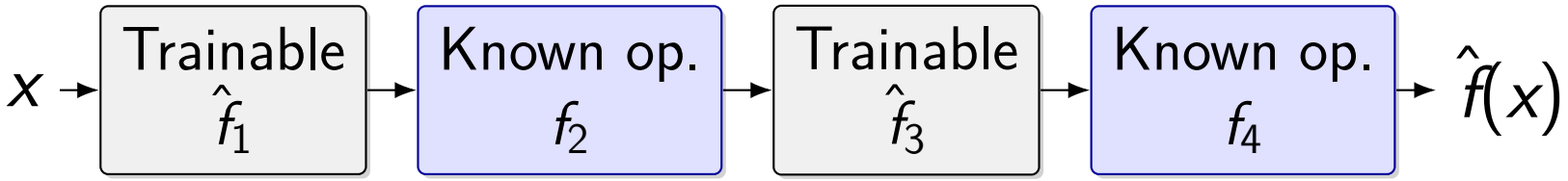
## Style variations 生成式人工智能






# Deep learning in medicine

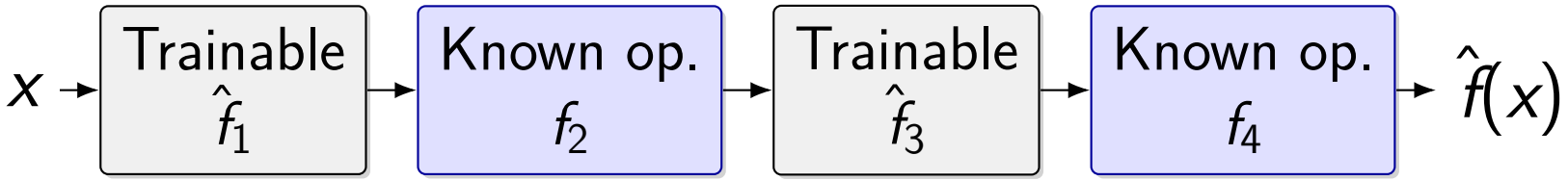
## Known operators 医学中的深度学习




Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

# Deep learning in medicine

## Known operators 医学中的深度学习




Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

**Theorem – known operators reduce the error bound**

For $f = f_n \circ \cdots \circ f_1$ approximated by $\hat{f} = \hat{f}_n \circ \cdots \circ \hat{f}_1$ with per-layer errors $\varepsilon_i$ and Lipschitz constants $\hat{L}_j$,

$$\varepsilon(\hat{f}) \leq \sum_{i=1}^{n} \left( \prod_{j=i+1}^{n} \hat{L}_j \right) \varepsilon_i .$$

Replacing $\hat{f}_k$ with the *known* $f_k$ forces $\varepsilon_k = 0$ and strictly tightens this bound.

Maier et al., *Nature Machine Intelligence*, 2019.

# Deep learning in medicine

## Recognition 医学中的深度学习







Deutscher Zukunftspreis – low-field MRI made possible by deep learning.

# What is agentic AI?

什么是智能体 AI?

# What is agentic AI?

## A simple question 什么是智能体 AI?

*“How do you drive a car?”*

*Thought for a few seconds...*

- A classic prompt to a large language model.
- After a short “thinking pause” the answer arrives as long-form running text.
- No tools, no actions – the model just *describes* how to drive.

# What is agentic AI?

## Long text answer of an LLM 什么是智能体 AI?

To drive a car, follow these steps:

- **Get in and adjust:** fasten your **seat belt**, set the **seat** and **mirrors** so you can see the road clearly.
- **Start the engine:** press the **brake pedal**, turn the key or push the start button. Make sure the gear selector is in **Park (P)** or **Neutral (N)**.
- **Engage a gear:** for an automatic, move the lever to **Drive (D)**; for a manual, press the **clutch** and select **1st gear**.
- **Release the parking brake** and slowly lift your foot off the brake. Gently press the **accelerator** to move forward.
- **Steer** smoothly with both hands and **check mirrors** regularly. Keep a safe distance from the vehicle in front.
- **To stop:** ease off the accelerator, press the brake, and bring the car to a complete stop. Shift to **Park** and apply the parking brake before turning off the engine.

*Always observe traffic rules and adapt your driving to road and weather.*

$\Rightarrow$ **Lots of words, but no *action*.**

# What is agentic AI?

## From language model to agent 什么是智能体 AI?



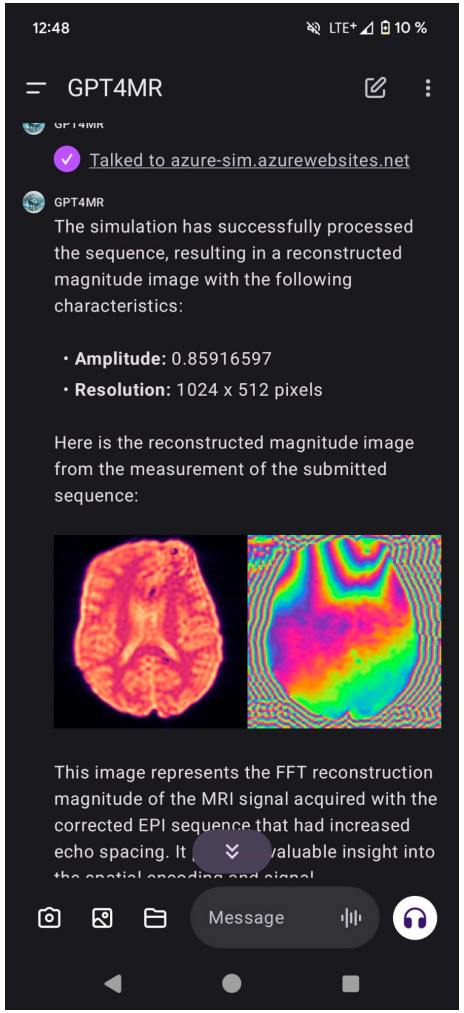
Moritz Zaiss

# What is agentic AI?

## From language model to agent 什么是智能体 AI?



Moritz Zaiss




Zaiss et al., *GPT4MR*, 2024.

# What is agentic AI?

## Task for the models 什么是智能体 AI?

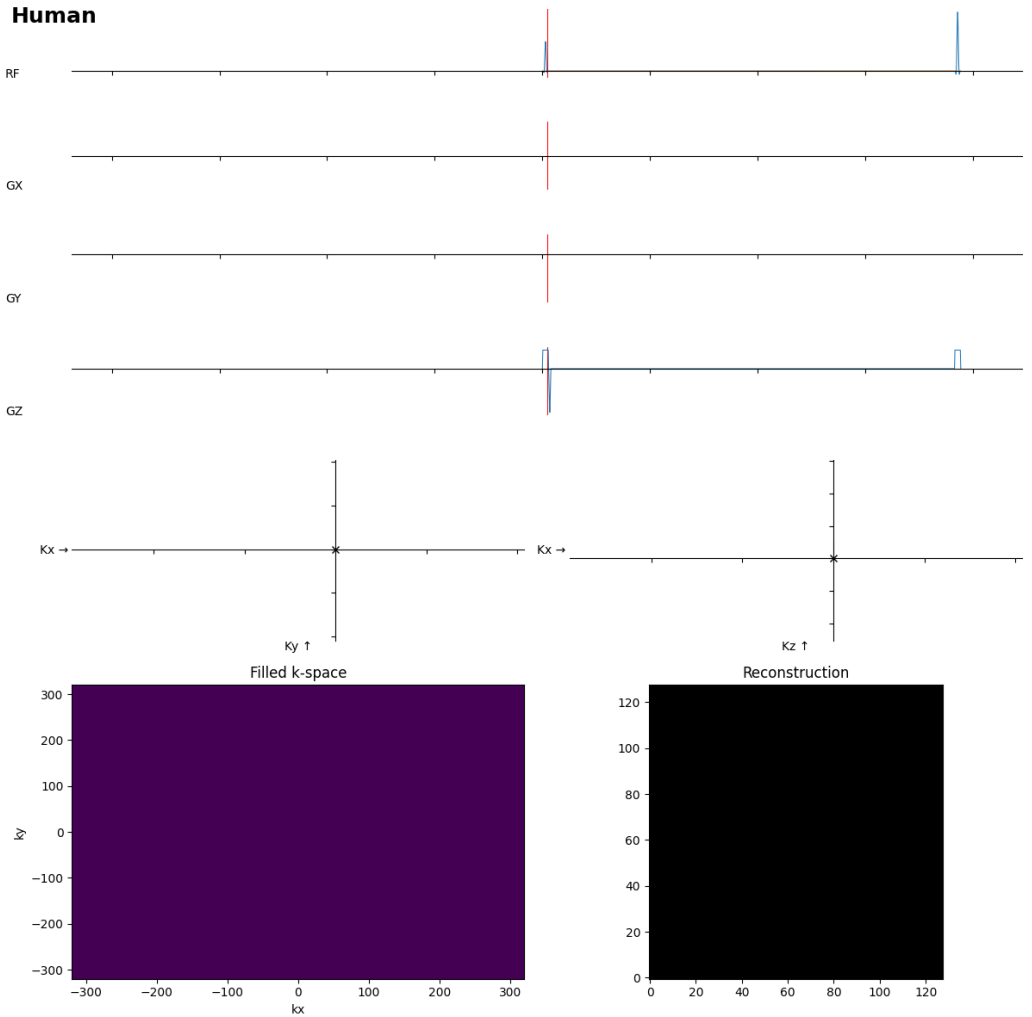
*"Code a spin echo EPI (64×64, fov = (0.2, 0.2, 0.08) m, TE = 150 ms)!"*

- **Spin-echo EPI** – an advanced MRI sequence used for diffusion and functional imaging.
- Three models receive the *same* natural-language instruction and must produce runnable Python code.
- Contestants:

| **Human** | **GPT-4o** | **GROK-3-think** |
|---|---|---|
| reference | OpenAI | xAI |

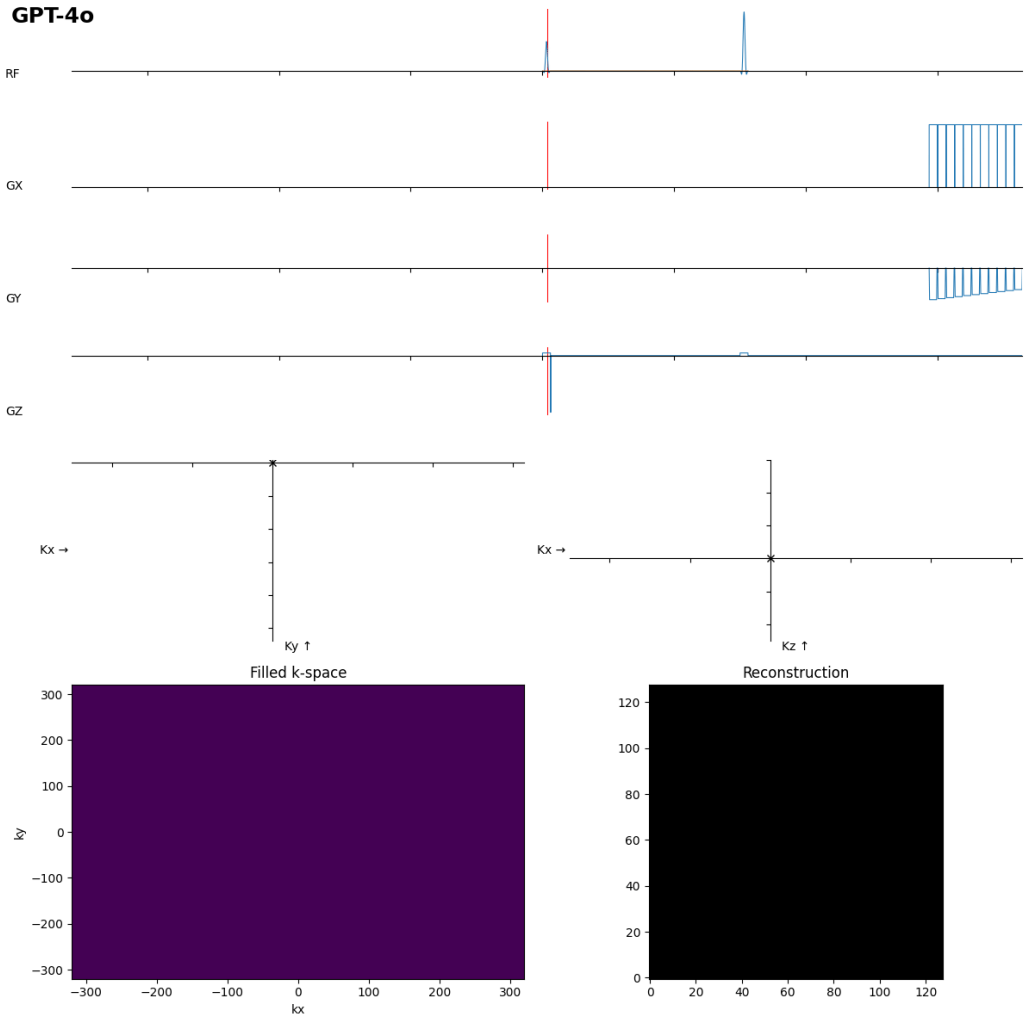
# What is agentic AI?

## Human baseline 什么是智能体 AI?



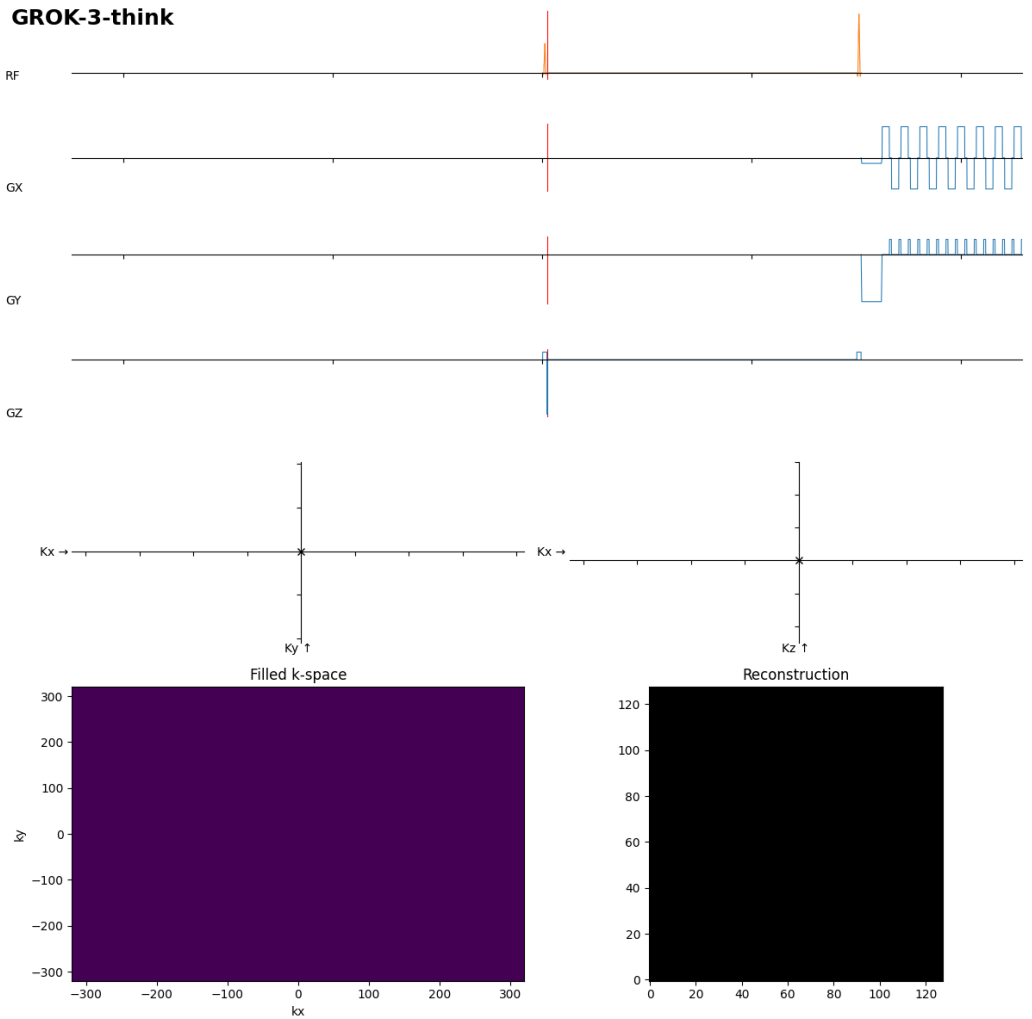
# What is agentic AI?

## GPT-4o attempt 什么是智能体 AI?



# What is agentic AI?

## GROK-3-think attempt 什么是智能体 AI?

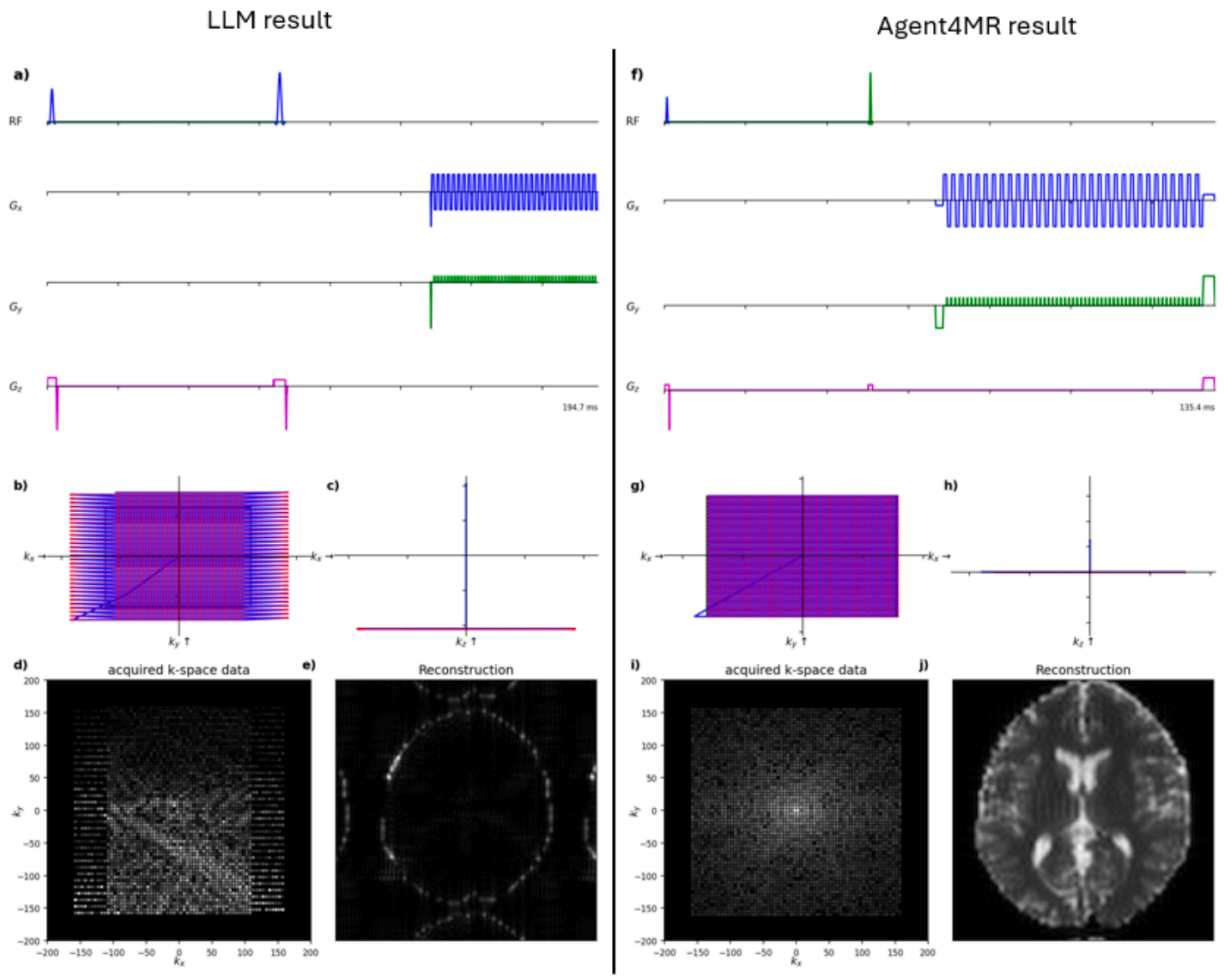


# How about real agentic AI?

真正的智能体 AI 又如何?

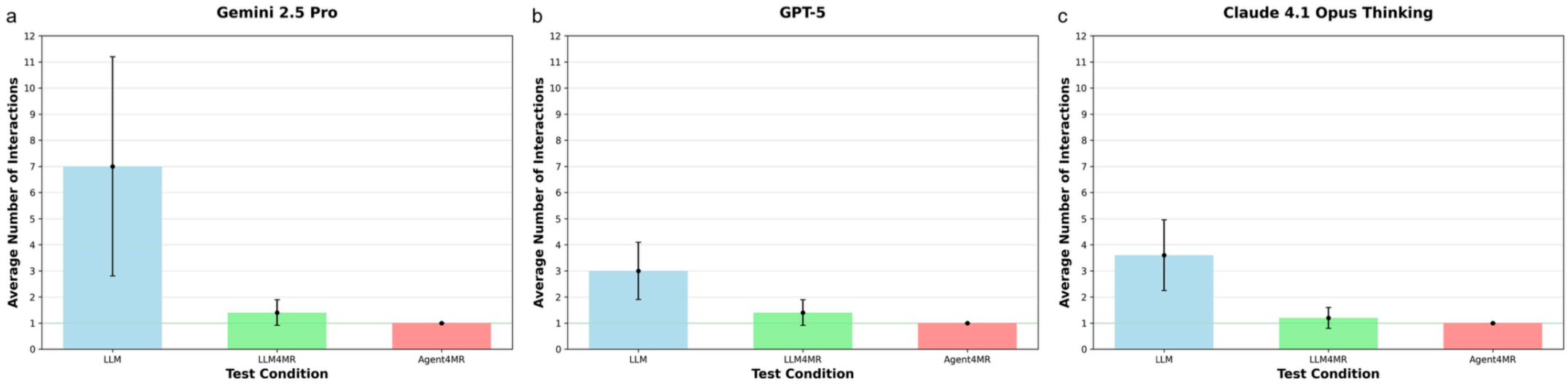
# What is agentic AI?

## LLM vs. Agent4MR 什么是智能体 AI?



# What is agentic AI?

## Interactions per model 什么是智能体 AI?



# What is agentic AI?

## Cost comparison 什么是智能体 AI?

Table 1: User interactions N, time, token usage, and estimated cost until a satisfactory sequence. Human baseline: recently trained EPI-naive student, 2.5 h, 5 interactions.

| Configuration Agent4MR | N | Time [min] | Tokens | Cost [€] [≈] |
|---|---|---|---|---|
| Gemini 2.5 Pro | 1 | 4.8 | 87k | 0.12 |
| GPT-5 | 1 | 15 | 520k | 0.48 |
| Claude 4.1 Opus | 1 | 10.2 | 1162k | 10.00 |
| Human | 5 | 150 | — | 37.50 |

# What is agentic AI?

## Iterative progress 什么是智能体 AI?

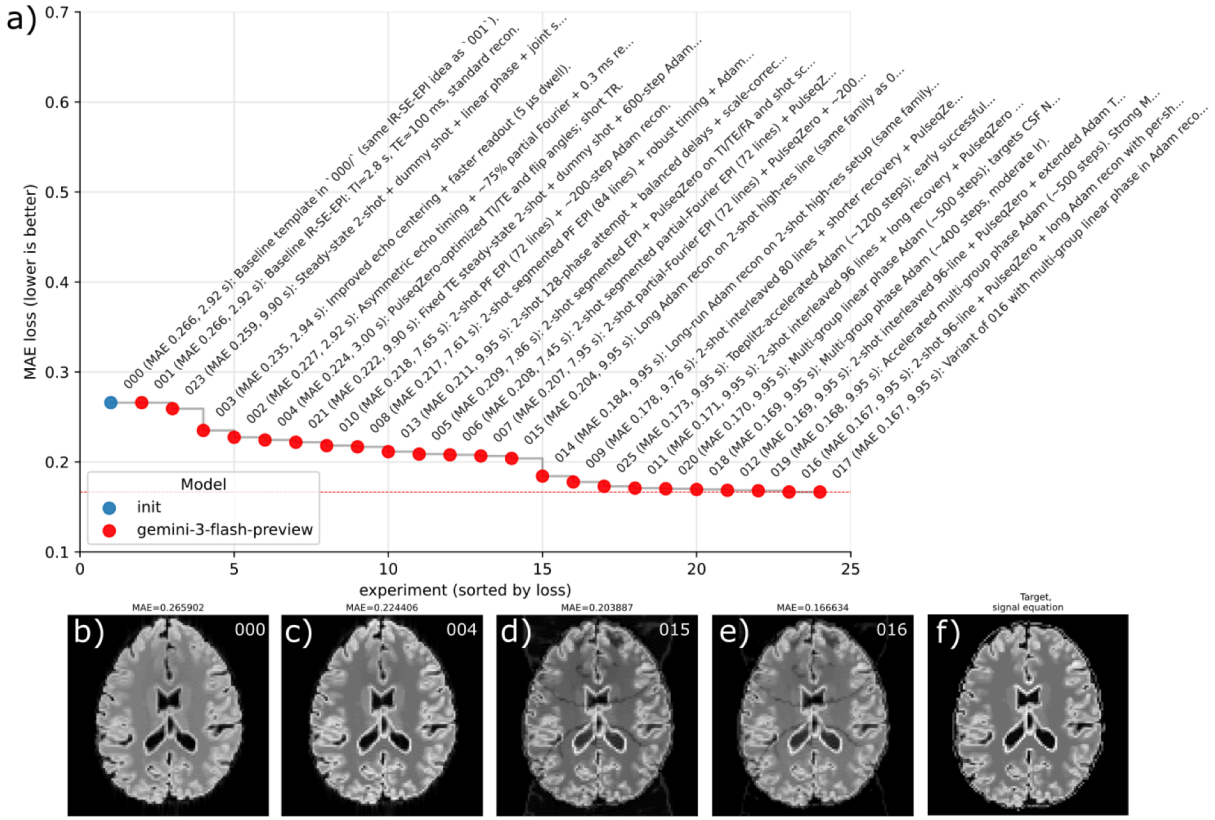


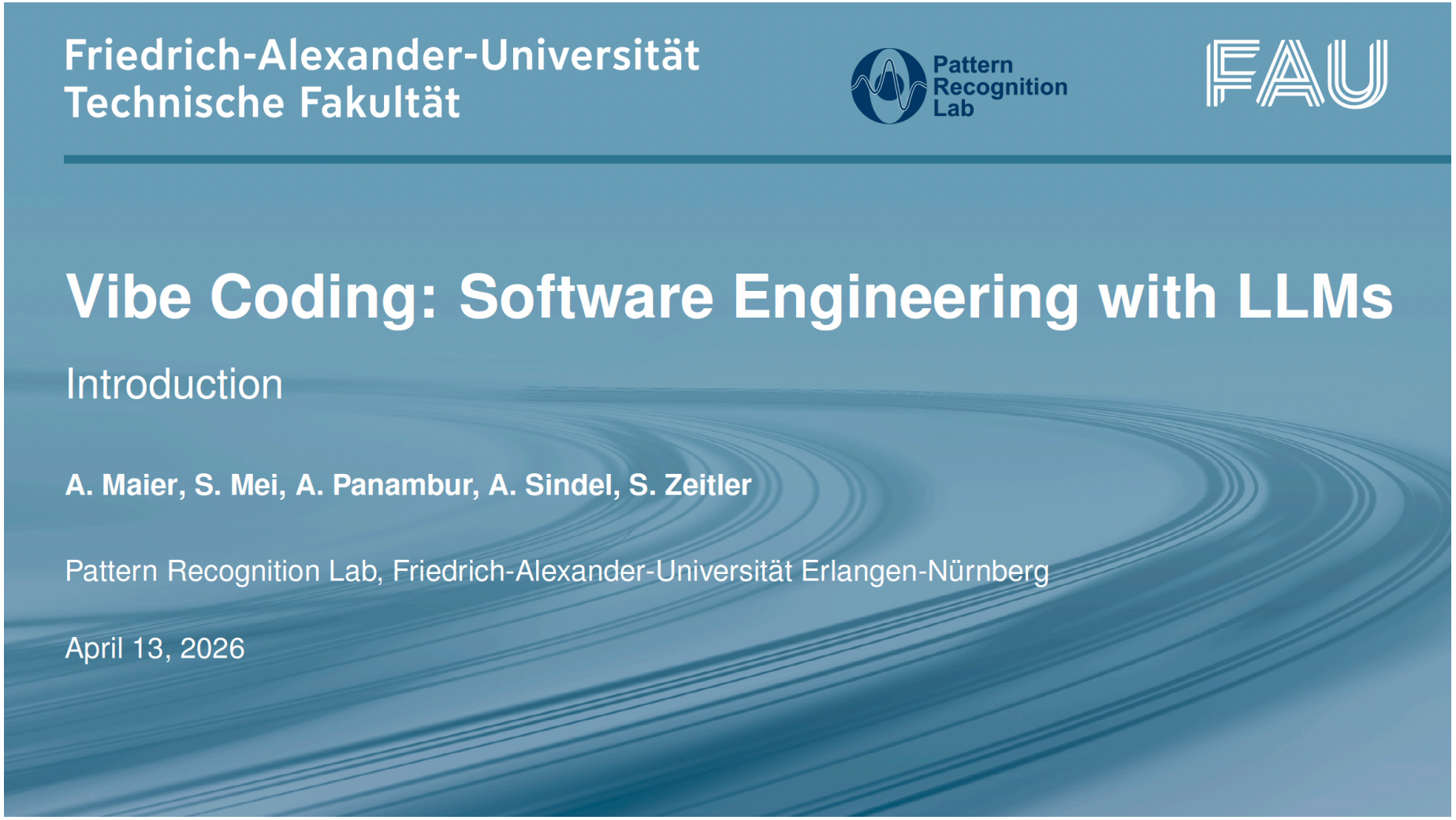

Figure 5: IR-SE-EPI MAE leaderboard progress. Autonomous agents iteratively improved the IR-SE-EPI pipeline (sequence parameters, reconstruction, post-processing). Experiments are ranked by MAE loss (worst to best). The staircase shows improvement from baseline (∼0.2659) to best (∼0.1666) through multi-window filtering, multi-shot introduction, phase correction, and differentiable TI/TE optimization.

Table 2: Normalized Token Usage and API Costs in EUR (n=25 events per model)

| Gemini Model | Total Tokens [Mio] | Tokens per Exp. [Mio] | Cost [€] |
|---|---|---|---|
| 2.5 Flash | 25.456 | 1.018 | 2.43 |
| 3 Flash | 136.340 | 5.453 | 12.85 |
| 3.1 Pro | 72.039 | 2.881 | 59.72 |
| **Total** | **233.836** | **9.353** | **75.00** |

# What is agentic AI?

## Vibe coding 什么是智能体 AI?



# What is agentic AI?

## Advanced AI-assisted techniques 什么是智能体 AI?

### Advanced AI-Assisted Techniques

**Claude Skills** [3]:

- Shareable workflow automation (YAML + Markdown)
- Team-wide best practices as code
- Deployment, testing, code review
- Available at `github.com/anthropics/skills`

**Model Context Protocol (MCP)**:

- Standardized tool integration
- Connect AI to databases, APIs, files
- Growing ecosystem of plugins

**Other Advanced Patterns:**

- **Agentic workflows**: Multi-step autonomous tasks
- **Tool-augmented coding**: AI uses external tools
- **Vision support**: Screenshot-driven debugging

**Key Benefit**

These techniques enable AI to perform complex, multi-step workflows beyond simple code generation.

to a programmer interface for an agent.

Pattern Recognition Lab A. Maier et al. Vibe Coding: Software Engineering with LLMs – Introduction December 2, 2025 11/27

# What is agentic AI?

## Markdown-driven quality gates 什么是智能体 AI?

**Geek Box 13.3: Markdown-driven quality gates: when agents run the production checklist**

A key insight from agentic software engineering is that quality criteria, which used to require human inspection, can now be expressed as structured markdown instructions that an AI agent executes autonomously. The agent reads the checklist, runs the specified checks, collects all findings, and presents them for human review—combining the thoroughness of automated scanning with the judgment-level assessments previously exclusive to human reviewers.

**Example: a production checklist for book chapters.** This book uses a markdown-based production checklist that an AI coding agent runs before each chapter is finalized. A condensed excerpt:

```
# Production Checklist for Chapter Review
The agent should: 1. Detect all problems first.
2. Present a PASS/FAIL table. 3. Ask the user
which to fix. 4. Apply only selected fixes.

## Checks to Run
### Cross-References
- All figures/tables/geek boxes referenced in text.
### Accessibility
- All figures have descriptive alt-text.
### Layout
- No "Float too large" or overfull hbox warnings.
### Acronyms
- No duplicate acronym expansions.
```

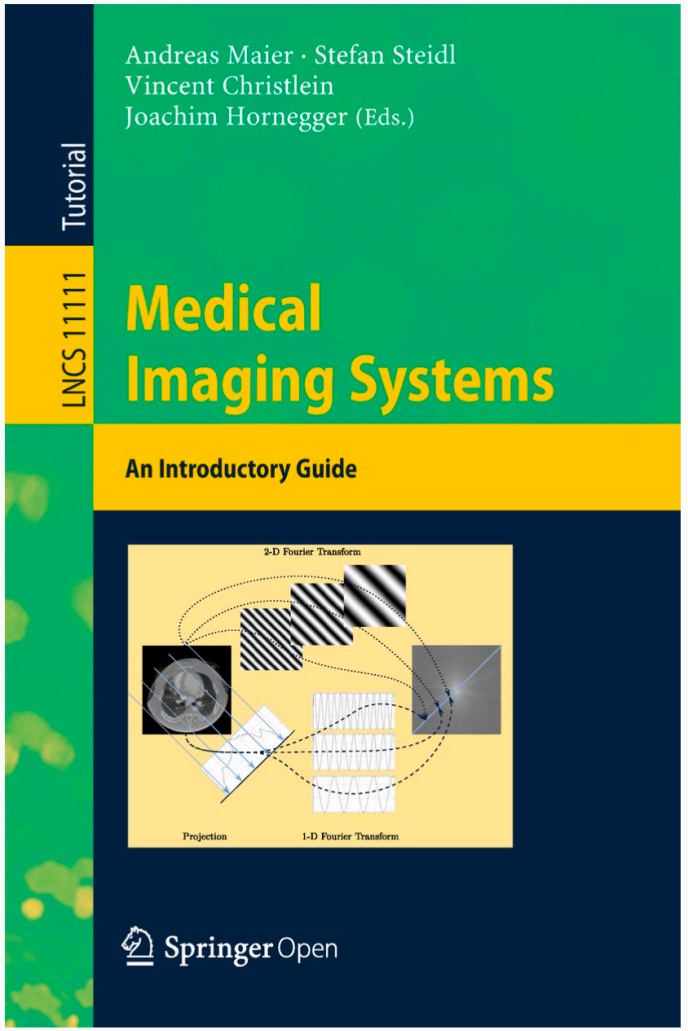
Several of these checks—alt-text quality, duplicate acronym detection, caption descriptiveness—require understanding the *meaning* of text, not just its syntax. An AI agent can evaluate whether alt-text actually describes what a figure shows or whether a manually spelled-out acronym conflicts with a `\gls{}` expansion nearby. These are assessments no simple script can perform reliably.

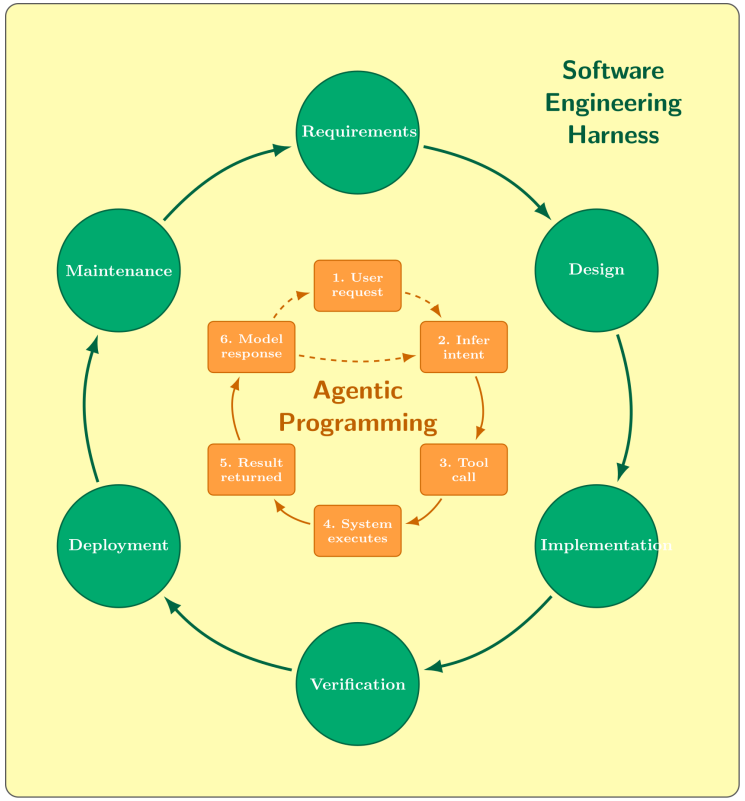
**The detect-then-ask workflow.** The checklist specifies a four-step process: detect all issues, present a PASS/FAIL summary, ask the user which fixes to apply, and then execute only approved changes. This separation of detection from correction mirrors the role-separation principle from Chapter 12: the agent that finds problems should not unilaterally fix them. The human decides which findings are genuine issues and which are acceptable.

This pattern generalizes beyond book production. Any domain with explicit quality criteria—code review checklists, regulatory compliance, accessibility audits, documentation reviews—can be encoded as markdown and executed by an agent. The quality gate becomes a reusable, version-controlled artifact rather than tacit knowledge in a reviewer's head.

# What is agentic AI?

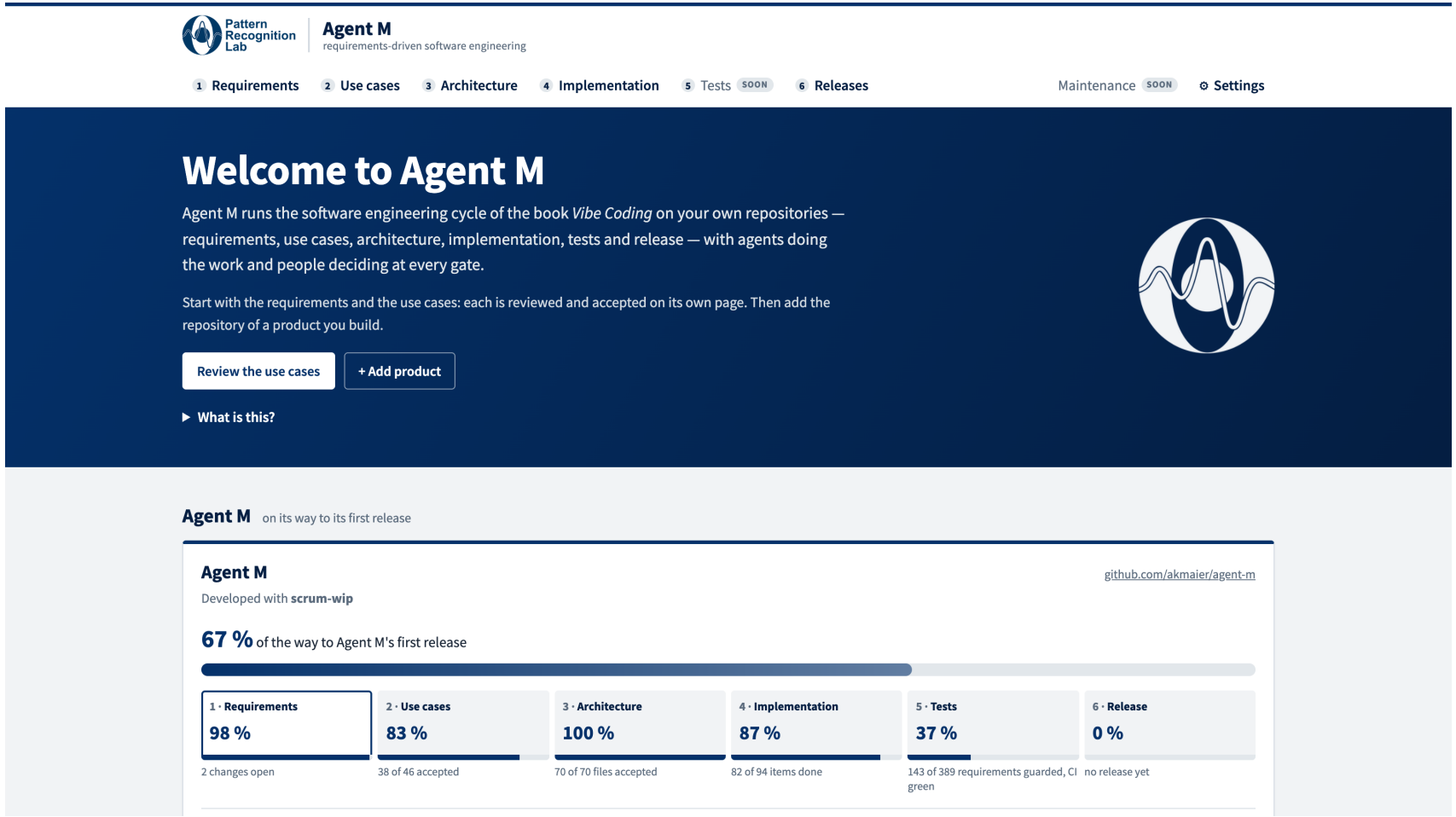
## Textbook 什么是智能体 AI?



# What is agentic AI?

## Agent M – companion tool to *Vibe Coding* 什么是智能体 AI?




akmaier.github.io/agent-m

# What does it mean for science?

这对科学意味着什么?

# Almost everything is easy now...

## China and AI 如今几乎一切都变得容易……



# Almost everything is easy now...

## Conference shares 如今几乎一切都变得容易……

| Conference | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|---|---|---|---|
| CVPR | ~30% | 32% | 35% | 35% | 39% | 39.2% | ~40% | ~38% | ~39% | ~40% |
| ICCV/ECCV | ~30% | 30% | 32% | 35% | ~37% | ~40% | ~40% | ~38% | ~40% | ~40% |
| ICML | ~10% | 12% | 15% | 18% | ~20% | ~22% | ~25% | ~28% | ~30% | ~30% |
| NeurIPS | 8% | 10% | 12% | 15% | *~15%* | ~18% | ~20% | ~25% | ~28% | ~30% |
| AAAI | 15% | 18% | 22% | 28% | ~35% | 50% | ~50% | ~45% | ~48% | ~50% |

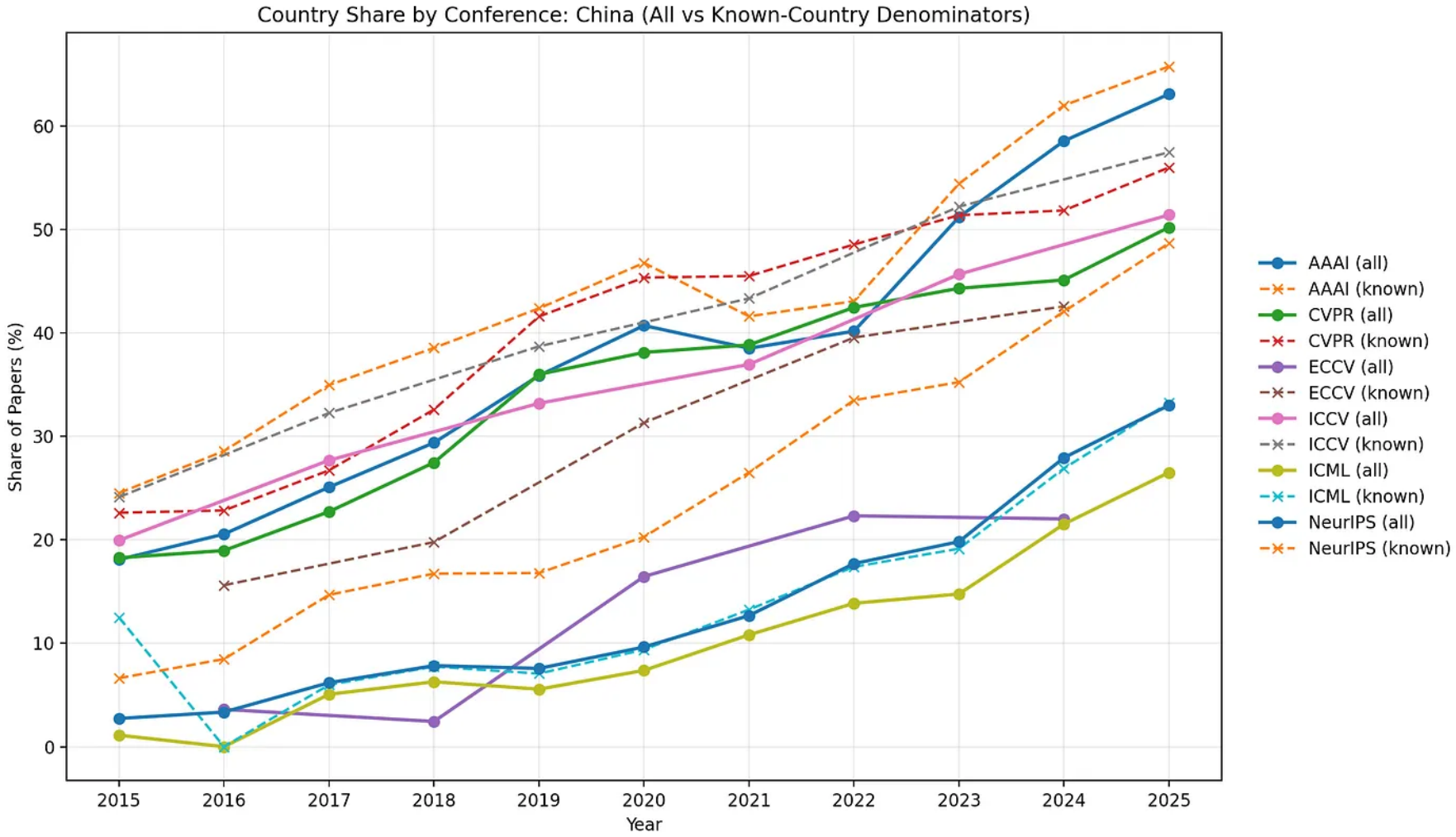
# Almost everything is easy now...

## ConferenceStats workflow 如今几乎一切都变得容易……



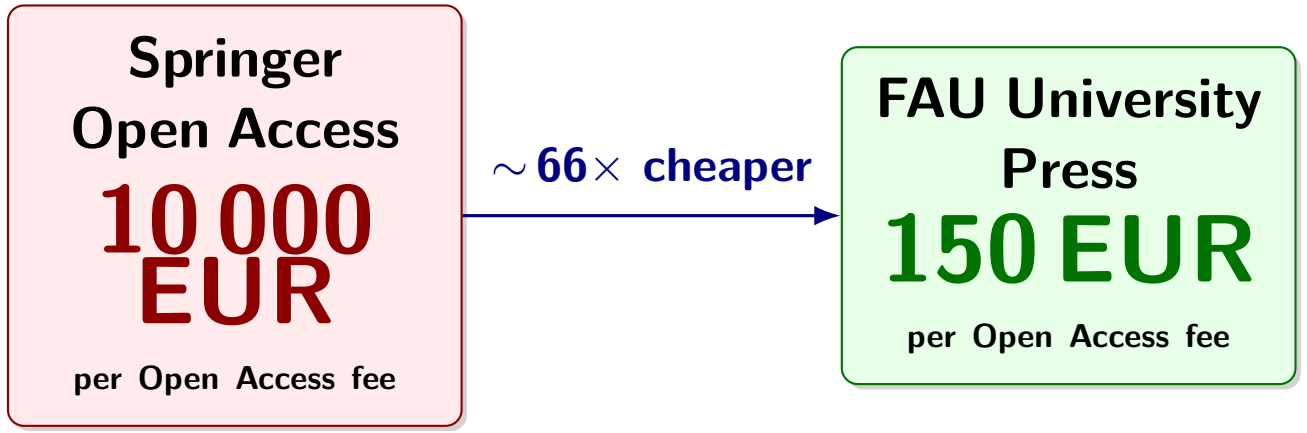
# Almost everything is easy now...

## Conference contributions from China 如今几乎一切都变得容易……



# Publishing, Disrupted

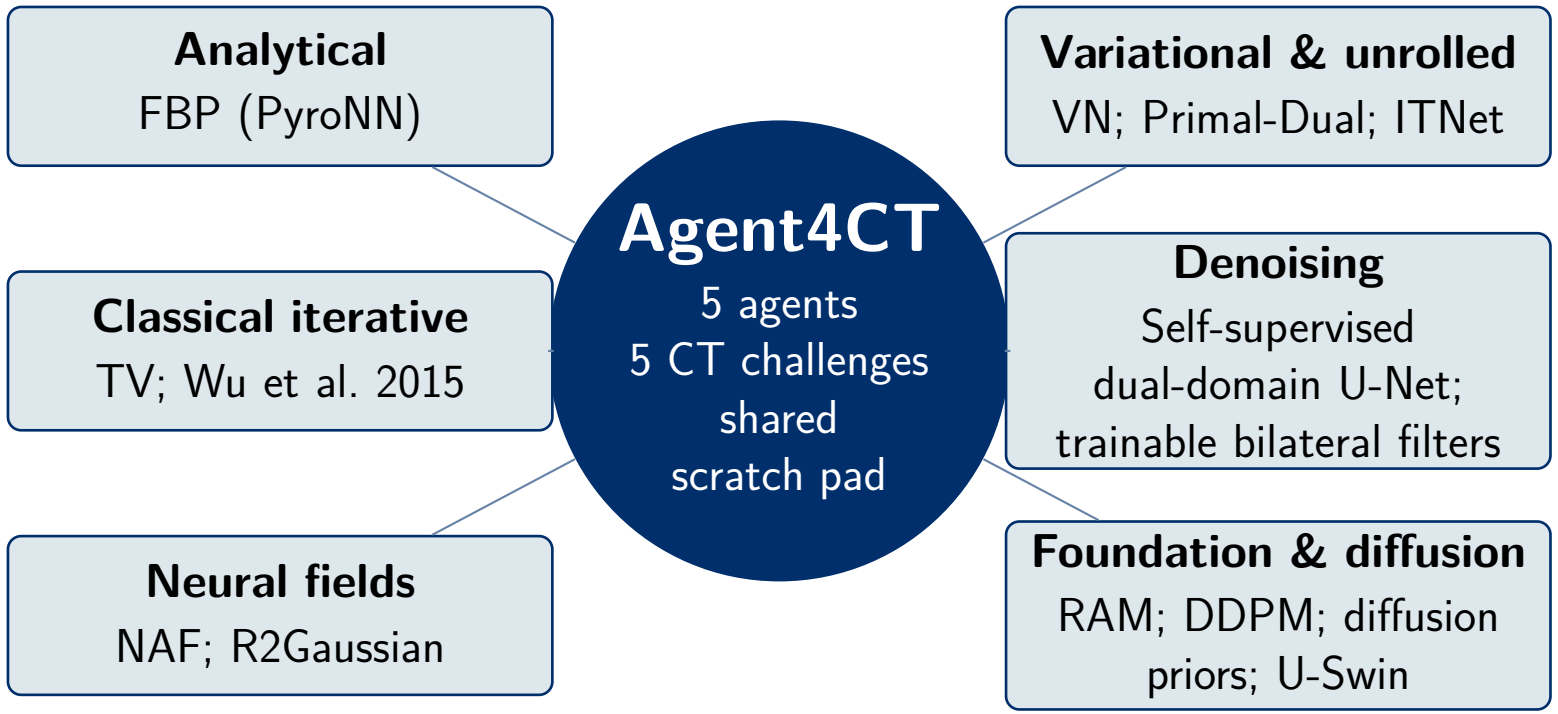
## Open Access for the price of a pizza party 出版业的变革




- **106** submissions → **93** accepted → one Open Access volume.
- Idea to printed book: **three months**, including **peer review**.

# Agent4CT

## All deep-learning CT methods, in one weekend 智能体复现 CT 方法




Classical re-implementation

$\approx$ **1 month**

per method – 15 methods

↓

Agent4CT

**one weekend**

all of them

the work of **15–20**

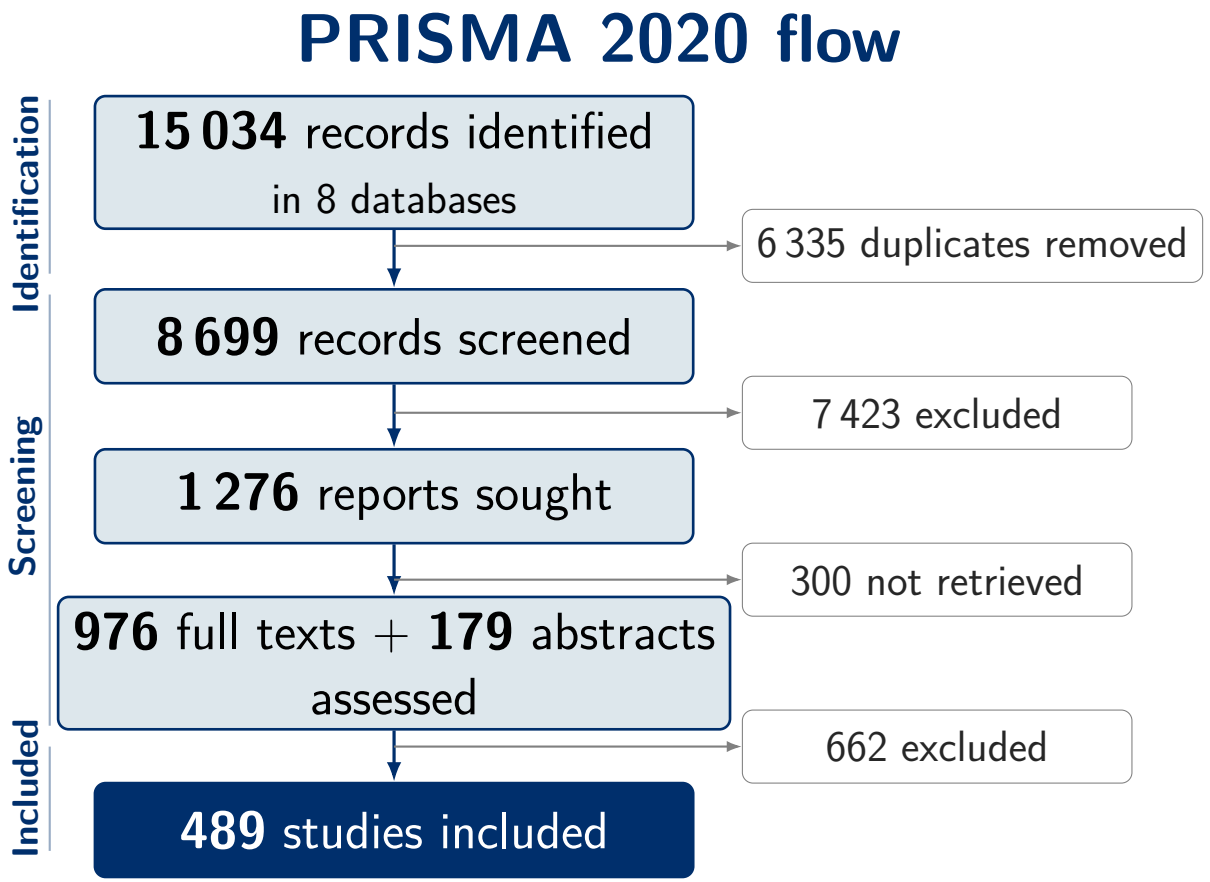
**PhD theses**

(re-implementation only)

`github.com/akmaier/Agent4CT`

# Living PRISMA Reviews

## Systematic reviews at agent speed 活体 PRISMA 系统综述




Writing time

**~~5 months~~ → 7 days**

- **PRISMA 2020** end to end: search, screening, extraction by a three-LLM panel; authors validate.
- **Automatic monthly update**: a *living* review.
- First use case: **robotic active matter**.

`akmaier.github.io/ActiveMatterLivingReview`

*Thanks to Siming Bayer for the idea!*

# Speed of Agentic AI

## Wall-clock from idea to working artefact　智能体 AI 的速度

**3 h** Reproduce a paper + code, find new insights, draft a follow-up paper.
arxiv.org/abs/2605.02620

**1 d** Reproduce a paper *without* its code or data.
arxiv.org/abs/2605.12772

**3 d** Scrape 200 study programs + ship a RAG chatbot.
github.com/akmaier/IfCampoKnew

**1 w** Reimplement an entire ATARI VCS emulator – bit-exact, 16 k lines of code.
github.com/akmaier/UnderstandingVCS

# Science at Agent Speed

## Opportunities and risks 智能体速度下的科学

### Opportunities

- Months of work in hours.
- Reproduce papers without code.
- Open Access for 150 EUR.
- Coding for every discipline.

### Risks

- Plausible is not correct.
- Hallucinated references.
- AI-text detectors beaten in 3 h.
- Sponsored bias in 12 LLMs.

Maier et al., arXiv:2605.02620 (style detector) ▪ Maier et al., arXiv:2605.12772 (sponsored recommendations)

# What This Means for Your Research

## Five rules for working with agents 这对你的研究意味着什么

1. Be the verifier, not the typist.
2. Reproduce before you build.
3. Keep a git-backed lab journal.
4. Write quality gates, not just prompts.
5. Disclose AI use – own the final word.

# References

## 参考文献


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- **A. Maier** et al., *Learning with Known Operators Reduces Maximum Error Bounds*, *Nat. Mach. Intell.*, 1:373–380, 2019.
- M. Zaiss, J. R. Rajput, H. N. Dang, V. Golkov, D. Cremers, F. Knoll and **A. Maier**, *Exploring GPT-4 as MR sequence and reconstruction programming assistant: GPT4MR*, BVM Workshop, pp. 94–99, Springer, 2024.
- M. Zaiss, A. Aly, J. Endres, T. Dornstetter, S. Weinmüller and **A. Maier**, *Agentic MR sequence development: leveraging LLMs with MR skills for automatic physics-informed sequence development*, arXiv:2604.13282, 2026 (Agent4MR).
- Deutscher Zukunftspreis (low-field MRI): https://www.deutscher-zukunftspreis.de
- Agent M – companion tool to the book *Vibe Coding*: https://github.com/akmaier/agent-m
- Conference statistics tooling: https://github.com/akmaier/ConferenceStats
- Agent4CT – agentic re-implementation of deep-learning CT methods: https://github.com/akmaier/Agent4CT
- **A. Maier**, M. Zaiss and S. Bayer, *Beating the Style Detector: Three Hours of Agentic Research on the AI-Text Arms Race*, arXiv:2605.02620, 2026.
- **A. Maier**, J. Sopa, G. G. Sahin, P. Perez-Toro and S. Bayer, *Just Ask for a Table: A Thirty-Token User Prompt Defeats Sponsored Recommendations in Twelve LLMs*, arXiv:2605.12772, 2026.
- *IfCampoKnew* – scraper for 200 study programs and an AI RAG chatbot: https://github.com/akmaier/IfCampoKnew
- *UnderstandingVCS* – agent-reimplemented ATARI VCS emulator: https://github.com/akmaier/UnderstandingVCS
- **A. K. Maier**, S. Bayer and W. Han, *Combining rater fusion and time fusion for coronary centerline extraction from single-plane angiography*, SPIE Medical Imaging 2027 (accepted); code: https://github.com/akmaier/VesselROVER
- **A. Maier**, S. Bayer et al., *A Living Systematic Review of Robotic Active Matter*: https://akmaier.github.io/ActiveMatterLivingReview

# Thank you! Questions?

## 谢谢！欢迎提问

Slides on GitHub 幻灯片见 GitHub



github.com/akmaier/AIMed2026Slides/tree/shanghai-sino-euro-2026

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